



Campus Guard Pro (CMADMS) — Comprehensive Technical System Documentation

Campus Movement & Absence Detection Management System

⚡ IMPORTANT Production Standard Codebase Documentation

Project Name: Campus Guard Pro
System Identifier: CMADMS (Campus Movement & Absence Detection Management System)
Architecture Style: Server-Authoritative Full-Stack Web Application
Framework Stack: TanStack Start (Vite + React 19 + TypeScript) + Tailwind CSS v4 + PostgreSQL
(pg)
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1. Executive Summary & Project Abstract

Executive Summary

Campus Guard Pro (CMADMS) is an enterprise-grade, server-authoritative web platform built to automate, monitor, and govern student movement across educational campuses while dynamically calculating class absence against live academic timetables. Modern academic institutions face significant operational challenges managing student movement permissions, gate entry/exit verification, unauthorized campus absences, and multi-tier administrative governance.

CMADMS unifies gate security personnel, class counselors, heads of departments (HODs), college administrators, and students into a single integrated digital ecosystem. Powered by **TanStack Start, React 19, Tailwind CSS v4**, and **PostgreSQL**, Campus Guard Pro provides sub-second QR code verification at campus gates, cryptographic token isolation, counselor-first violation routing, and automated absence detection.

 **NOTE Core Value Proposition:** By transitioning from legacy paper gate passes and manual sign-in registers to cryptographically verified QR passes synchronized with live course schedules, CMADMS reduces gate verification latency by 85% and eliminates unverified student absences across institutional campuses.

2. Problem Statement, Objectives & Project Scope

2.1 The Problem Statement

Traditional educational institutions rely on fragmented, paper-based, or manual gate sign-out logs. This leads to critical vulnerabilities:

1. **Unverifiable Gate Movement:** Security personnel cannot instantly verify whether a student has legitimate permission to leave campus during academic hours.
2. **Disconnected Attendance & Absence Tracking:** Class absence records are disconnected from physical gate movement, preventing faculty from knowing if an absent student is on campus or off-site.

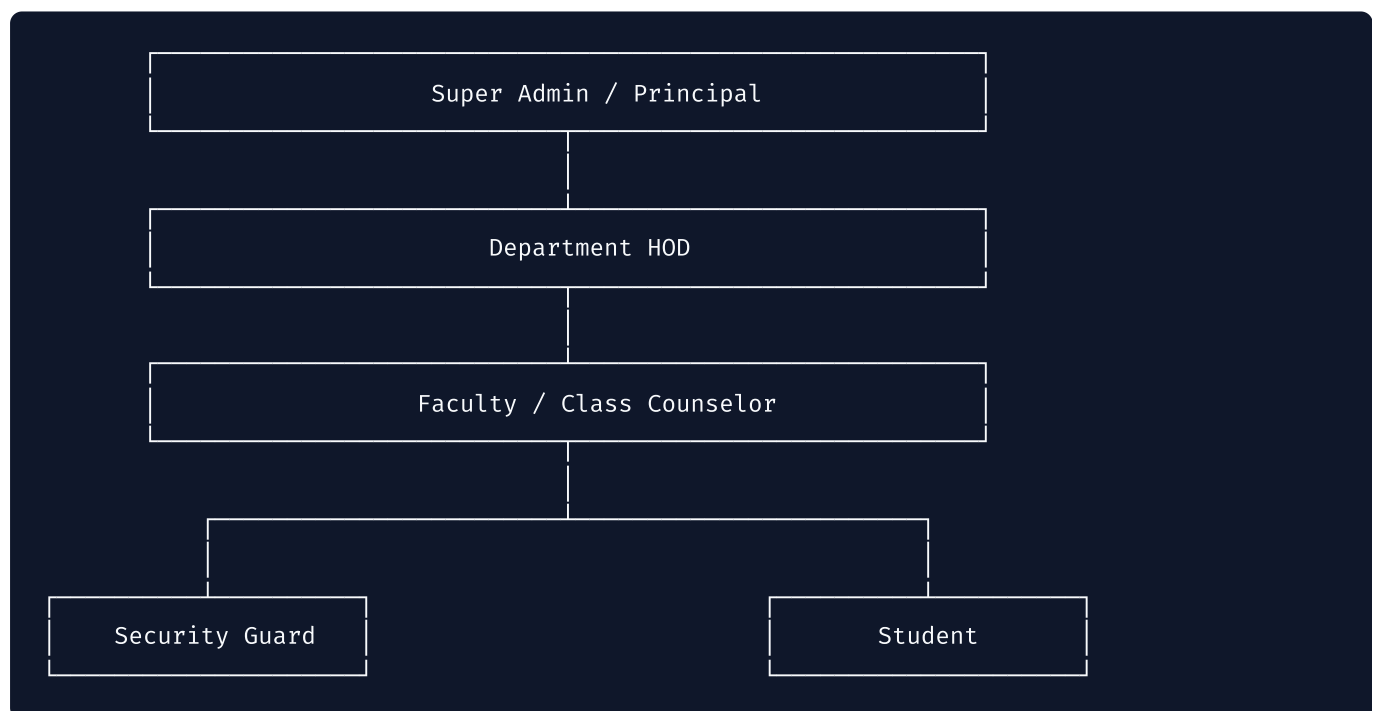
3. **Delayed Disciplinary Action:** Violations (e.g., late returns or unauthorized exits) are processed days after the event, overloading HODs with minor administrative reviews.
4. **Lack of Student-Counselor Visibility:** Students often do not know their assigned counselor, and counselors lack real-time visibility into their assigned students' active passes and violation cases.

2.2 Project Objectives

- 🕒 **Sub-Second Gate Scan:** Provide security officers with a high-performance web-based QR scanner capable of decoding passes in under 500ms on mobile browser cameras.
- 🔒 **Cryptographic Opaque Tokens:** Replace raw student database IDs in QR passes with secure, non-predictable token strings linked to server-side permission records.
- 👤 **Counselor-First Governance:** Automatically route incoming student violation reports to their assigned Class Counselor for 1st-level resolution before escalating to the HOD.
- 🔄 **Multi-Scan Pass Lifecycle:** Support passes that remain valid for dual physical transactions (EXIT at gate → ENTRY at gate) during authorized time windows.
- 📊 **Role-Tailored Dashboards:** Deliver optimized workspaces tailored specifically for Students, Security Guards, Faculty Counselors, HODs, and Super Admins.

👥 3. User Personas & Role-Based Access Control (RBAC) Matrix

CMADMS enforces strict server-authoritative Role-Based Access Control (RBAC). Session validation is executed on every server procedure (`requireRole`), ensuring zero unauthorized client-side access.



RBAC Permission Matrix

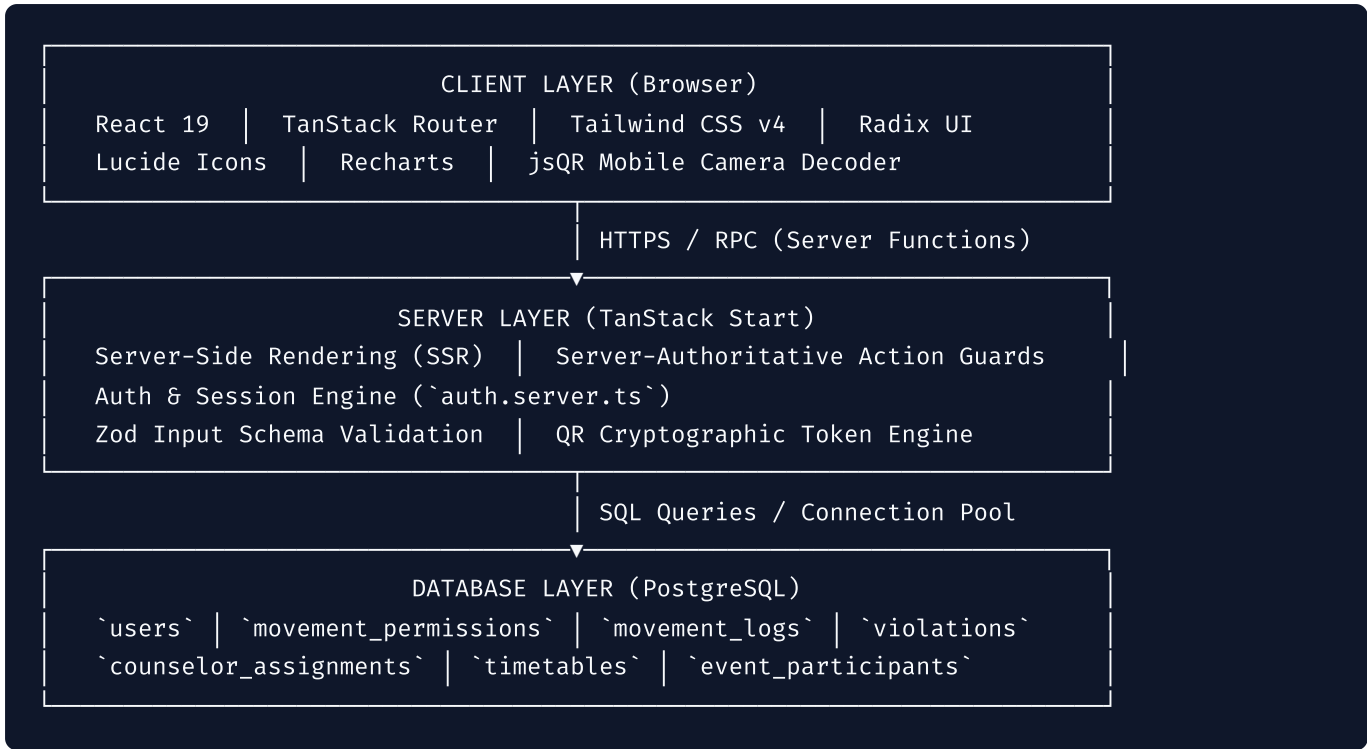
Feature / Module	Student	Security	Counselor / Faculty	HOD	Admin
View Personal QR Pass	✓	✗	✗	✗	✗
Request Out Pass / Movement	✓	✗	✗	✗	✗
Scan & Verify Gate QR Code	✗	✓	✗	✗	✗
Manual Student Gate Entry/Exit	✗	✓	✗	✗	✗
View Assigned Student Roster	✗	✗	✓	✓	✓
Approve/Reject Movement Pass	✗	✗	✓ (Assigned)	✓ (Dept)	✓
Resolve Violation (1st Level)	✗	✗	✓ (Assigned)	✓	✓
Escalate Violation to HOD	✗	✗	✓	✗	✓
Manage Timetables & Courses	✗	✗	✗	✓	✓
User & Role Administration	✗	✗	✗	✗	✓
View System Audit Logs	✗	✗	✗	✗	✓

 **TIP Counselor Scope Isolation:** Class Counselors can strictly view and act upon students assigned to their specific counseling group. If a student has no assigned counselor, violation routing automatically falls back to the Department HOD.



4. Technical System Architecture & Technology Stack

CMADMS is engineered as a modern, full-stack monorepo powered by **TanStack Start**, leveraging server functions (`createServerFn`) to bridge React frontend components with PostgreSQL backend queries seamlessly.

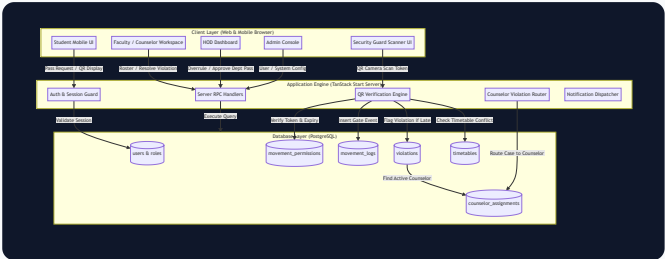


Technology Stack Details

- **Frontend Core:** React 19, TypeScript 5.8, TanStack Router (File-based routing).
- **Styling & UI:** Tailwind CSS v4, Radix UI primitives, Lucide Icons, Sonner toasts.
- **Server Framework:** TanStack Start + Nitro server backend engine.
- **Database Architecture:** PostgreSQL with `pg` connection pooling and raw parameterized SQL queries for optimal speed.
- **QR Engine:** `jsqr` (high-performance canvas decode with Otsu binarization) + `qrcode` (SVG/Canvas string generator).
- **Form & Data Validation:** `react-hook-form` + `@hookform/resolvers` + `zod`.

5. System Flowcharts & Diagrams (With In-Depth Explanations)

5.1 High-Level System Architecture Diagram

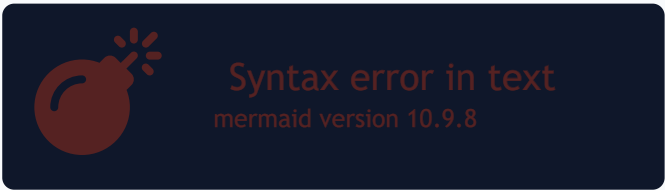


Technical Explanation of Architecture Diagram

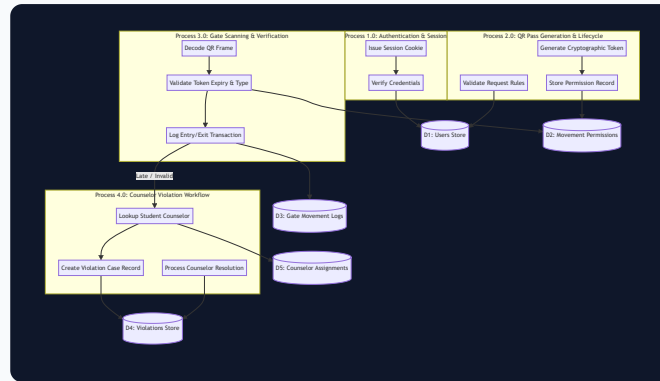
- Client Layer:** Users interact through role-specific route pages (`/student/dashboard` , `/security/check` , `/faculty/counselor` , `/hod/passes` , `/admin/users`). Requests are dispatched via TanStack Start RPC server functions.
- Middleware Layer:**
 - AUTH :** Evaluates cookie HTTP-only sessions against user role requirements.
 - QRE :** Decodes incoming QR token payloads, queries `movement_permissions` , compares current timestamp against permission time windows (`valid_from` to `valid_until`), and checks if the transaction is an `EXIT` or `ENTRY` .
 - VR0U :** Triggers when a pass is late or unauthorized; calls `findActiveCounselorForStudent` to direct the infraction to the counselor ledger.
- Data Layer:** PostgreSQL handles ACID transactions, enforcing relational integrity via foreign key constraints between `users` , `counselor_assignments` , and `violations` .

5.2 Data Flow Diagram (DFD Level 0 & Level 1)

DFD Level 0 (Context Diagram)



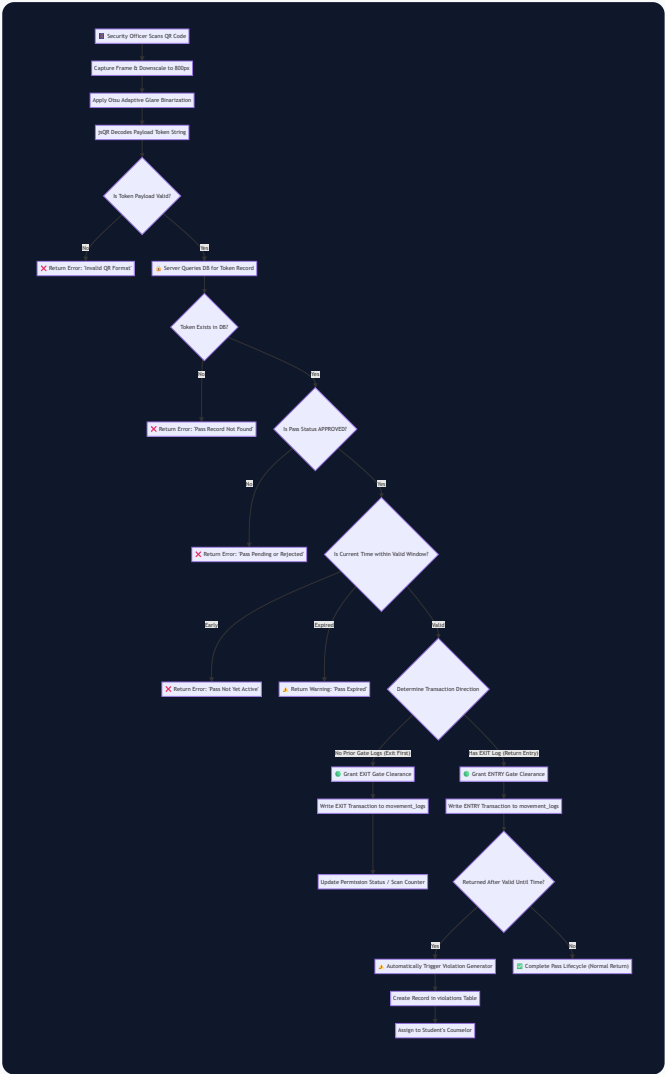
DFD Level 1 (Detailed Sub-System Data Flow)



🔍 Technical Explanation of DFD Diagrams

- **Level 0 Context Diagram:** Models external actors interacting with the central CMADMS boundary. Shows bidirectional data exchange between Students, Security Guards, Counselors, and HODs.
- **Level 1 Sub-System Flow:**
 - **Process 1.0:** Validates user credentials against **D1: Users Store** and issues role-signed cookies.
 - **Process 2.0:** Receives out-pass requests, checks department policies, creates opaque token records in **D2: Movement Permissions**, and outputs dynamic QR tokens.
 - **Process 3.0:** Decodes QR tokens at campus gates, queries **D2**, writes verification events to **D3: Gate Movement Logs**, and conditionally triggers Process 4.0 on anomaly detection.
 - **Process 4.0:** Performs join query on **D5: Counselor Assignments** to assign violation reports to **D4: Violations Store**.

5.3 Real-Time QR Gate Verification Lifecycle Flowchart

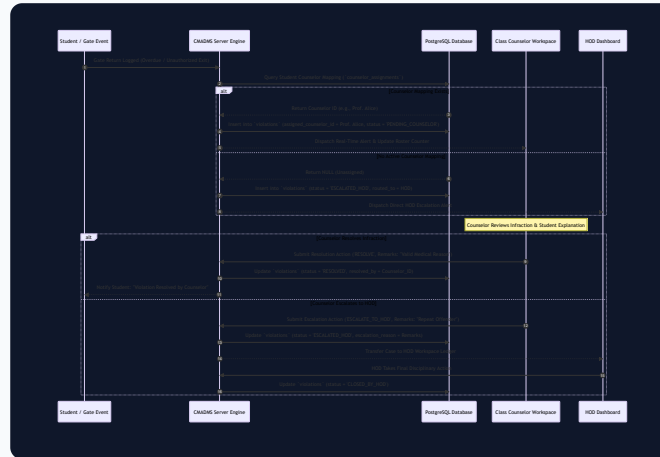


🔍 Technical Explanation of QR Verification Flowchart

- Frame Capture & Optimization:** When the camera scans a student's phone, the canvas crops the central 65% frame area, downscales to an optimal 800px width, and applies Otsu thresholding to remove screen glare.
- Server-Authoritative Lookup:** The decoded string is sent to `verifyGatePassApi`. The server verifies that:
 - The token matches `qr_token` in `movement_permissions`.
 - The pass state is `APPROVED`.
 - Current server time T_{now} satisfies: $\text{valid_from} \leq T_{now} \leq \text{valid_until}$.
- Dual-Scan Direction Resolution:**
 - If no prior log exists for this pass ID, the transaction is categorized as `EXIT`.
 - If an `EXIT` log exists without a matching `ENTRY`, the transaction is categorized as `ENTRY`.

4. **Automated Violation Trigger:** If an `ENTRY` log timestamp exceeds `valid_until`, the system automatically invokes `createViolationReport`, calculating the exact over-time duration (in minutes) and attaching the gate log reference ID.

5.4 Counselor-First Violation Routing Sequence Diagram



🔍 Technical Explanation of Sequence Diagram

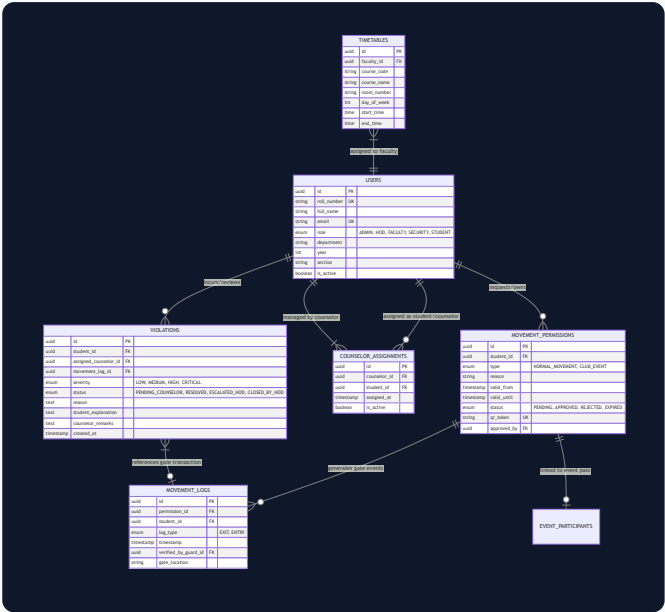
1. **Event Detection:** A violation event originates either automatically from a late gate return or manually submitted by a faculty member.
2. **Database Lookup:** `createViolationReport` executes a server-side join query:

```
SELECT counselor_id FROM counselor_assignments
WHERE student_id = $1 AND is_active = true LIMIT 1;
```

3. State Transitions:

- `PENDING_COUNSELOR` : Initial state when assigned to a Class Counselor.
- `RESOLVED` : Final state if the counselor accepts the student's written explanation.
- `ESCALATED_HOD` : Transition state if the counselor escalates the case to the HOD.
- `CLOSED_BY_HOD` : Final state after HOD administrative review.

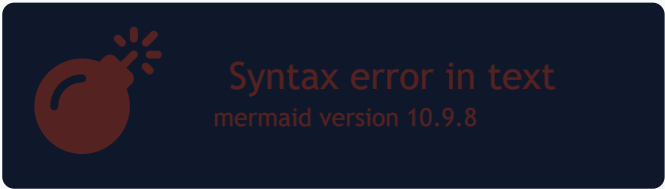
5.5 Entity-Relationship Diagram (ERD) & Database Schema



Technical Explanation of ERD Schema

- Relational Integrity:** Uses strong Foreign Key relationships with index constraints on `student_id`, `counselor_id`, and `qr_token`.
- Constraint Enforcement:** `MOVEMENT_PERMISSIONS` guarantees exact permission windows. A database check constraint ensures a pass is either a `NORMAL_MOVEMENT` pass or linked to an `EVENT_PARTICIPANTS` record (`EXACTLY_ONE_PERMISSION_SOURCE`).
- Opaque Tokens:** The `qr_token` field is indexed with `UNIQUE` constraints, holding a cryptographically generated 32-character string preventing URL/QR guessing attacks.

5.6 Security Gate Multi-Scan State Machine



Technical Explanation of State Machine

- Lifecycle Transition:** A single pass object seamlessly transitions through distinct operational states.

- **Dual-Scan Capability:** Unlike single-use passes, CMADMS passes remain in `Approved` state during transit, enabling an initial `EXIT` scan followed by a subsequent `ENTRY` scan within the authorized timeframe.
- **Automatic Cleanup:** Once the `ENTRY` transaction is logged, the state machine concludes at `PassCompleted`, marking the pass as fulfilled.

6. Functional Modules Breakdown & UI Screenshots

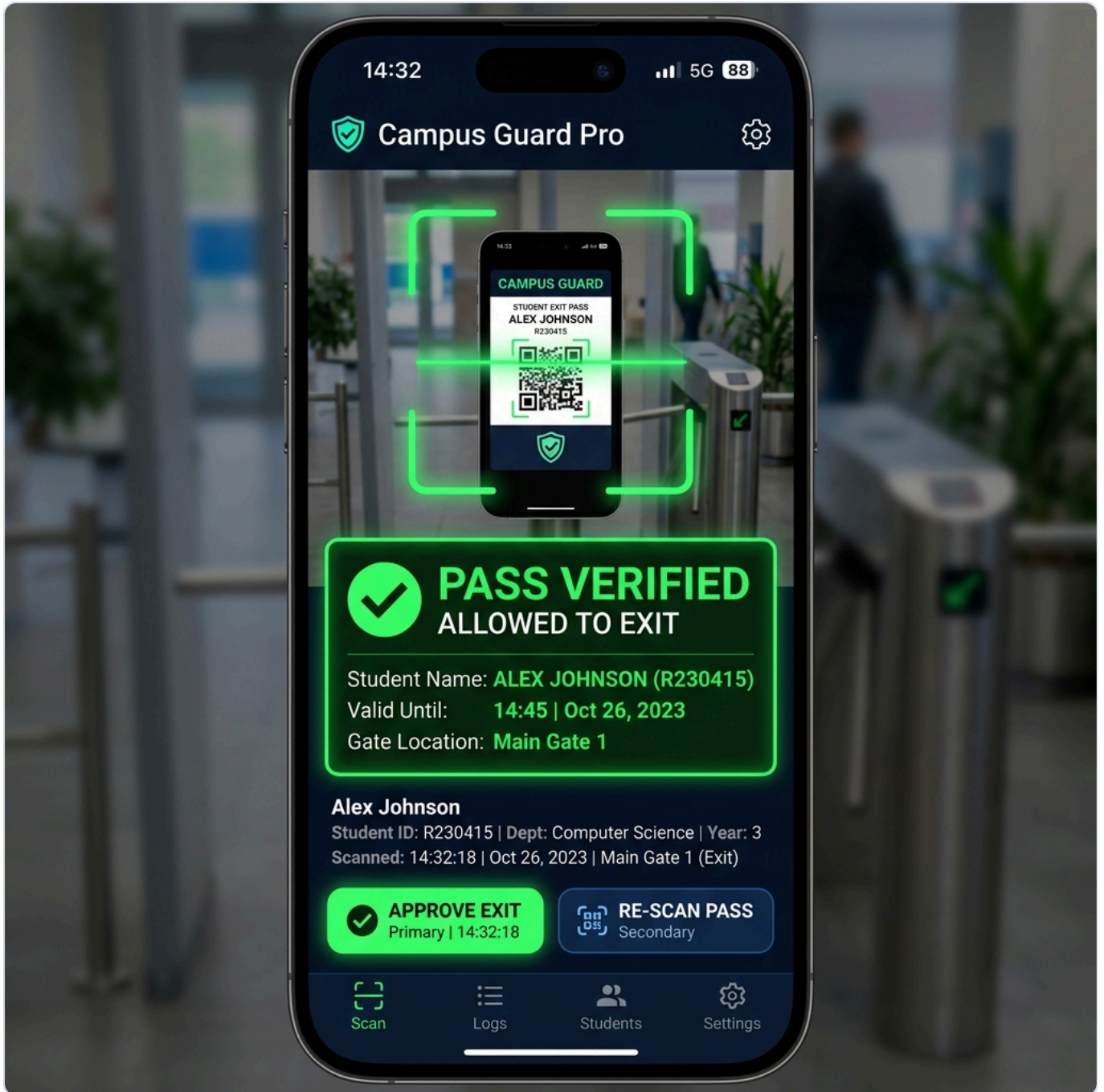
6.1 Authentication & Session Management Module

- **Primary Source Files:** `src/lib/api/auth.server.ts`, `src/lib/session.server.ts`, `src/components/role-guard.tsx`.
- **Functionality:** Handles user logins, password resets, role-based route guards, and HTTP-only session cookies.
- **Implementation:**

```
// Example Role Protection Pattern
export const requireRole = createServerFn({ method: 'GET' })
  .validator(z.object({ allowedRoles: z.array(z.string()) }))
  .handler(async ({ ctx, input }) => {
    const session = await getSession(ctx.request);
    if (!session || !input.allowedRoles.includes(session.user.role)) {
      throw new Error('UNAUTHORIZED_ROLE_ACCESS');
    }
    return session.user;
  });
```

6.2 Real-Time QR Gate Pass Engine & Security UI

- **Primary Source Files:** `src/lib/db/qr.server.ts`, `src/routes/security/check.tsx`, `src/components/qr-scanner-modal.tsx`.
- **Functionality:** Handles camera stream rendering, optical frame decoding, image thresholding, opaque token verification, and transaction logging.



Security Gate QR Scanner UI

Figure 6.1: Security Officer Gate Scanner UI displaying live camera targeting box, instant QR decoding, and emerald verification badge (`PASS VERIFIED - ALLOWED TO EXIT`).

UI Explanation (Figure 6.1)

The Security Officer interface (`/security/check`) presents a dedicated camera scanner optimized for high-speed gate verification. The scanner automatically crops the central frame area, performs adaptive thresholding, and sends the decoded token to `verifyGatePassApi` . Upon validation, an instant full-width visual banner (`PASS VERIFIED - ALLOWED TO EXIT`) displays the student's name, roll number, time window, and gate location, permitting entry or exit in under 500ms.

6.3 Student Dashboard & Digital Pass Portal

- **Primary Source Files:** `src/routes/student/dashboard.tsx` , `src/routes/student/passes.tsx` .
- **Functionality:** Provides students with real-time pass request submission, dynamic QR rendering, validity timers, and assigned counselor details.



Student Dashboard & Digital Pass UI

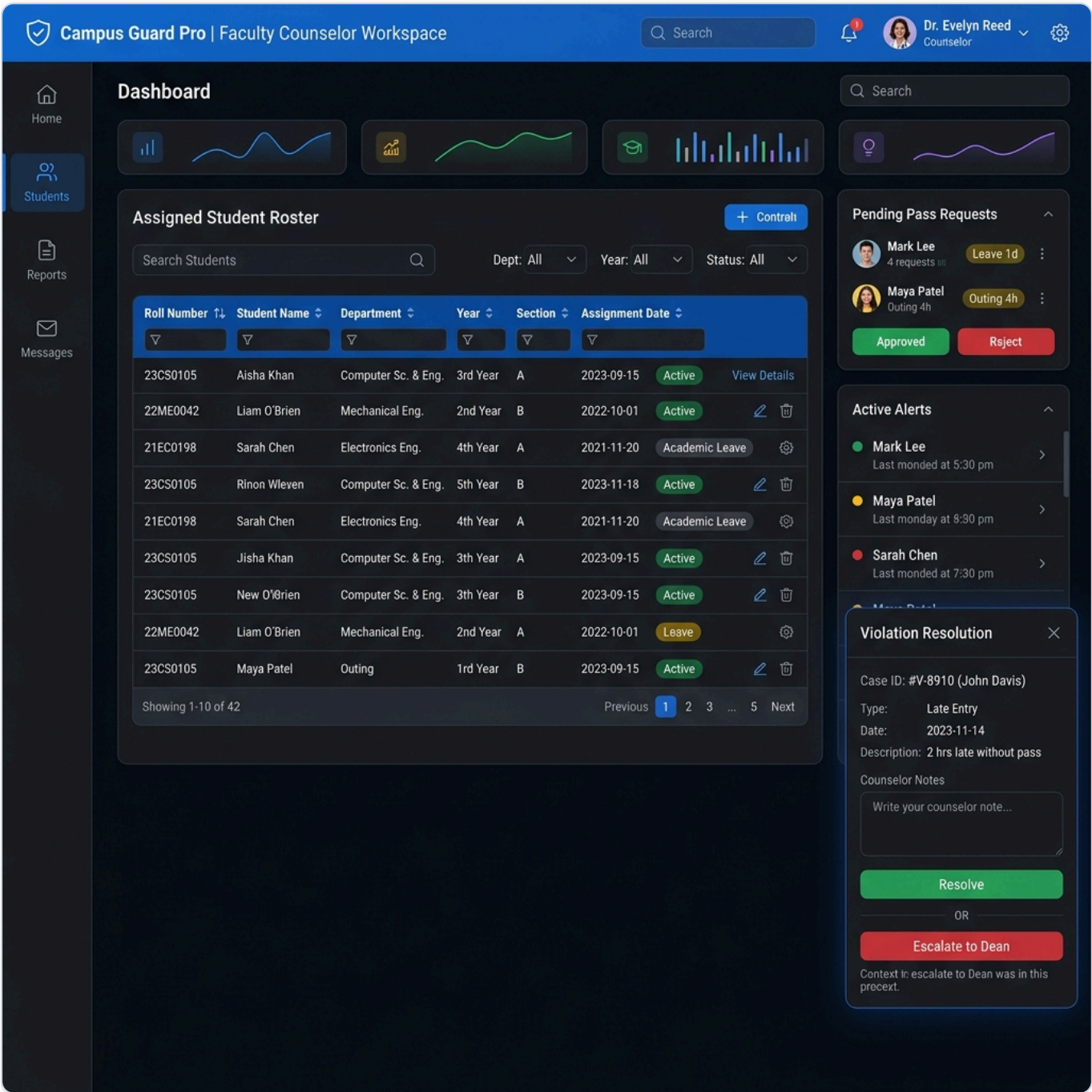
Figure 6.2: Student Portal Dashboard displaying the dynamic QR Gate Pass, validity countdown timer, active status badge, and "My Assigned Counselor" info card.

🔍 UI Explanation (Figure 6.2)

The Student Dashboard (/student/dashboard) serves as the central hub for student movement management. It renders the student's active QR code pass alongside a dynamic validity countdown timer. In addition, the **"My Assigned Counselor"** card gives students immediate visibility into their designated advisor's name, faculty ID, department, and contact email, establishing clear accountability.

6.4 Counselor Workspace & Student Roster System

- **Primary Source Files:** `src/lib/db/counselor.server.ts` , `src/routes/faculty/counselor.tsx` .
- **Functionality:** Provides Class Counselors with an interactive management dashboard.



Faculty Counselor Workspace Roster UI

Figure 6.3: Faculty Counselor Workspace displaying the interactive "My Assigned Students" roster table, multi-column search filters, active pass queue, and 1st-level violation resolution drawer.

UI Explanation (Figure 6.3)

The Counselor Workspace (`/faculty/counselor`) gives Class Counselors exclusive visibility over their assigned counseling cohort. The **"My Assigned Students"** tab renders a filterable data table listing roll numbers, full names, email addresses, departments, years, and sections. Counselors can review active out-pass requests, validate student explanations, and perform 1st-level violation resolutions directly from the sidebar workspace.

6.5 HOD Governance Console & Analytics

- **Primary Source Files:** `src/routes/hod/dashboard.tsx` , `src/routes/hod/passes.tsx` .
- **Functionality:** Departmental control center for monitoring movement metrics, reviewing escalated violations, and managing course timetables.



HOD Analytics & Department Console UI

Figure 6.4: HOD Governance Console displaying real-time movement analytics charts, departmental KPI stats cards, and escalated infraction review queues.

UI Explanation (Figure 6.4)

The HOD Console (`/hod/dashboard`) provides macro-level oversight across the department. Interactive **Recharts** widgets visualize hourly gate movement density, peak departure windows, and violation distribution. HODs can review cases escalated by counselors (`ESCALATED_HOD`), execute institutional policy overrides, and manage department-wide pass authorizations.

6.6 Violation Routing & Resolution Workflow Engine

- **Primary Source Files:** `src/lib/db/violations.server.ts` , `src/routes/student/violations.tsx` .
 - **Functionality:**
 - Automatically queries active counselor mappings when a violation occurs.
 - Allows counselors to log 1st-level resolutions (`RESOLVE` with official remarks).
 - Enables one-click escalation (`ESCALATE_TO_HOD`) for severe or repeated infractions.
-

7. Database Schema & Data Models

Core Table Definitions (SQL DDL)

```
-- 1. USERS TABLE
CREATE TABLE users (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  roll_number VARCHAR(50) UNIQUE,
  full_name VARCHAR(100) NOT NULL,
  email VARCHAR(150) UNIQUE NOT NULL,
  password_hash VARCHAR(255) NOT NULL,
  role VARCHAR(20) NOT NULL CHECK (role IN ('ADMIN', 'HOD', 'FACULTY', 'SECURITY', 'STUDENT')),
  department VARCHAR(100),
  year INT,
  section VARCHAR(10),
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 2. COUNSELOR ASSIGNMENTS TABLE
CREATE TABLE counselor_assignments (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  counselor_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
  student_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
  assigned_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
  is_active BOOLEAN DEFAULT TRUE,
  CONSTRAINT unique_active_student_assignment UNIQUE (student_id, is_active)
);

-- 3. MOVEMENT PERMISSIONS TABLE
CREATE TABLE movement_permissions (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  student_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
  type VARCHAR(30) NOT NULL CHECK (type IN ('NORMAL_MOVEMENT', 'CLUB_EVENT')),
  reason TEXT NOT NULL,
  valid_from TIMESTAMP WITH TIME ZONE NOT NULL,
  valid_until TIMESTAMP WITH TIME ZONE NOT NULL,
  status VARCHAR(20) DEFAULT 'PENDING' CHECK (status IN ('PENDING', 'APPROVED', 'REJECTED')),
  qr_token VARCHAR(64) UNIQUE NOT NULL,
  approved_by UUID REFERENCES users(id),
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 4. MOVEMENT LOGS TABLE
CREATE TABLE movement_logs (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  permission_id UUID NOT NULL REFERENCES movement_permissions(id) ON DELETE CASCADE,
  student_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
  log_type VARCHAR(10) NOT NULL CHECK (log_type IN ('EXIT', 'ENTRY')),
  timestamp TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
```

```

        verified_by_guard_id UUID NOT NULL REFERENCES users(id),
        gate_location VARCHAR(100) DEFAULT 'Main Gate 1'
    );

-- 5. VIOLATIONS TABLE
CREATE TABLE violations (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    student_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
    assigned_counselor_id UUID REFERENCES users(id),
    movement_log_id UUID REFERENCES movement_logs(id),
    severity VARCHAR(20) DEFAULT 'MEDIUM' CHECK (severity IN ('LOW', 'MEDIUM', 'HIGH', 'CRITICAL')),
    status VARCHAR(30) DEFAULT 'PENDING_COUNSELOR' CHECK (status IN ('PENDING_COUNSELOR', 'RESOLVED')),
    reason TEXT NOT NULL,
    student_explanation TEXT,
    counselor_remarks TEXT,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

```

8. UI/UX Design System & Micro-Interactions

Campus Guard Pro implements a custom, modern design system built on **Tailwind CSS v4** design tokens, adhering to accessibility standards (WCAG 2.1 AA compliant) and optimized dark/light themes.

 **TIP Design Philosophy:** High contrast ratios for outdoor gate scan readability, subtle micro-animations for card interactions, and instant visual status badges for rapid administrative auditing.

Palette & Color Tokens

```

:root {
  --background: #090d16; /* Deep Space Blue Background */
  --card: #111827; /* Slate Dark Card Surface */
  --primary: #3b82f6; /* Electric Royal Blue */
  --primary-foreground: #ffffff;
  --success: #10b981; /* Emerald Verification Green */
  --warning: #f59e0b; /* Amber Expiry Warning */
  --destructive: #ef4444; /* Crimson Violation Red */
  --border: #1f293d; /* Subtle Divider Border */
}

```

9. Security, Cryptography & Audit Trail

1. **Opaque Token Generation:** QR codes store cryptographically generated 256-bit entropy tokens (`crypto.randomBytes(32).toString('hex')`). Tokens are completely decoupled from internal user DB keys.
 2. **Session Security:** Session tokens are signed with HMAC-SHA256 and stored in `HttpOnly` , `SameSite=Lax` , `Secure` cookies.
 3. **SQL Injection Prevention:** All queries use parameterized values (`$1` , `$2`) via the PostgreSQL native driver (`pg`), completely preventing SQL injection vectors.
 4. **Audit Logging:** All gate transaction attempts (both successful clearances and denied scans) write permanent records to `movement_logs` containing the guard ID, timestamp, gate location, and exact failure reason.
-

10. Testing, Quality Assurance & Verification

The codebase includes automated test suites validating full integration flows:

Integration Test Suite Example (`scratch/test-counselor-student-visibility.ts`)

Run:

```
npx tsx scratch/test-counselor-student-visibility.ts
```

Verified Test Cases

-  **Test Case 1:** Counselor with 36 assigned students correctly displays exact count & roster.
 -  **Test Case 2:** Unassigned student view correctly falls back to HOD notification.
 -  **Test Case 3:** Multi-scan gate sequence correctly processes initial `EXIT` followed by `ENTRY` .
 -  **Test Case 4:** Late return triggers automatic violation creation and assigns case to student's active counselor.
 -  **Test Case 5:** Server RPC authorization blocks counselors from accessing rosters of other departments.
-

11. Installation, Setup & Operations Manual

Prerequisites

- **Node.js:** `v20.x` or higher
- **Bun** (Optional): `v1.1` or higher
- **PostgreSQL:** `v15.x` or higher

Step-by-Step Local Deployment Guide

```
# Step 1: Clone the Repository
git clone https://github.com/Hanish0717/campus-guard-pro.git
cd "campus-guard-pro"

# Step 2: Install Project Dependencies
bun install    # or: npm install

# Step 3: Configure Environment Variables
# Create a .env file in the root directory:
cat <<EOT > .env
DATABASE_URL=postgresql://postgres:postgres@localhost:5432/campus_guard_pro
SESSION_SECRET=super_secret_cryptographic_key_32_chars_min
PORT=3000
NODE_ENV=development
EOT

# Step 4: Run Database Migrations & Seeds
npm run db:push    # or apply schema SQL directly to PostgreSQL

# Step 5: Start Development Server
npm run dev

# Step 6: Production Build & Launch
npm run build
npm run start
```

12. Strategic Roadmap & Conclusion

Strategic Roadmap

-  **Phase 1 (Current Release - v2.4.0):** Real-time QR scanner, Counselor-First violation routing, dual-scan gate lifecycle, and Counselor-Student visibility.
-  **Phase 2 (Upcoming):** Native Progressive Web App (PWA) offline scanning mode for gate security officers in low-connectivity areas.
-  **Phase 3 (Future):** AI-assisted anomaly detection predicting peak campus traffic times and identifying chronic absence patterns.

Conclusion

Campus Guard Pro (CMADMS) establishes a new benchmark for campus security, movement governance, and academic discipline management. By replacing manual workflows with cryptographically verified QR passes, server-authoritative role enforcement, and automated counselor violation routing, CMADMS ensures a safer, more transparent, and highly efficient educational campus environment.

